```
from flask import Flask, request, redirect, render_template_string
import sqlite3
import os

app = Flask(__name__)

DATABASE = "students.db"

HTML = """
<!DOCTYPE html>
<html>
<head>
  <title>Student Management System</title>
  <meta name="viewport" content="width=device-width, initial-scale=1">

  <style>
    body {
      margin: 0;
      font-family: Arial, sans-serif;
      background: #f1f5f9;
      color: #1e293b;
    }

    header {
      background: #172554;
      color: white;
      padding: 22px;
      text-align: center;
    }

    header h1 {
      margin: 0 0 8px;
    }

    .container {
      max-width: 1000px;
      margin: 25px auto;
      padding: 15px;
    }

    .cards {
      display: grid;
      grid-template-columns: repeat(auto-fit, minmax(180px, 1fr));
      gap: 15px;
    }

    .card, .panel {
      background: white;
      padding: 20px;
```

```
    border-radius: 12px;
    box-shadow: 0 2px 8px #00000012;
    margin-bottom: 20px;
}

.card h2 {
    color: #2563eb;
    margin-bottom: 5px;
}

input {
    box-sizing: border-box;
    width: 100%;
    padding: 12px;
    margin: 7px 0 12px;
    border: 1px solid #cbd5e1;
    border-radius: 7px;
}

button {
    background: #2563eb;
    color: white;
    border: 0;
    padding: 11px 17px;
    border-radius: 7px;
    cursor: pointer;
}

.delete {
    background: #dc2626;
}

.search {
    display: flex;
    gap: 8px;
}

.search input {
    flex: 1;
}

.table-wrap {
    overflow-x: auto;
}

table {
    width: 100%;
    border-collapse: collapse;
    background: white;
}
```

```
        th, td {
            text-align: left;
            padding: 12px;
            border-bottom: 1px solid #e2e8f0;
        }

        th {
            background: #eff6ff;
        }

        footer {
            text-align: center;
            padding: 20px;
            color: #64748b;
        }
    </style>
</head>

<body>

<header>
    <h1>Student Management System</h1>
    <p>Navsahyadri College of Engineering</p>
    <p>Data Structure Mini Project</p>
</header>

<div class="container">

    <div class="cards">
        <div class="card">
            <p>Total Students</p>
            <h2>{{ students|length }}</h2>
        </div>

        <div class="card">
            <p>Project Subject</p>
            <h2>Data Structure</h2>
        </div>

        <div class="card">
            <p>Developed By</p>
            <h2>Chaitanya Chobhe</h2>
        </div>
    </div>

    <div class="panel">
        <h2>Add New Student</h2>

        <form action="/add" method="POST">
```

```
      <label>Roll Number</label>
      <input type="number" name="roll" required>

      <label>Student Name</label>
      <input type="text" name="name" required>

      <label>Course</label>
      <input type="text" name="course" required>

      <label>Email</label>
      <input type="email" name="email">

      <button type="submit">Add Student</button>
    </form>
  </div>

  <div class="panel">
    <h2>Search Students</h2>

    <form action="/" method="GET" class="search">
      <input type="text" name="q"
             placeholder="Search name or roll number"
             value="{{ query }}">
      <button type="submit">Search</button>
    </form>
  </div>

  <div class="panel">
    <h2>Student Records</h2>

    <div class="table-wrap">
      <table>
        <tr>
          <th>Roll No.</th>
          <th>Name</th>
          <th>Course</th>
          <th>Email</th>
          <th>Action</th>
        </tr>

        {% for student in students %}
        <tr>
          <td>{{ student[0] }}</td>
          <td>{{ student[1] }}</td>
          <td>{{ student[2] }}</td>
          <td>{{ student[3] }}</td>
          <td>
            <form action="/delete/{{ student[0] }}"
                  method="POST"
                  onsubmit="return confirm('Delete this student?')">
```

```
                    <button class="delete" type="submit">
                        Delete
                    </button>
                </form>
            </td>
        </tr>
        {% else %}
        <tr>
            <td colspan="5">No student records found.</td>
        </tr>
        {% endfor %}
    </table>
</div>
</div>

</div>

<footer>
    Student Management System | Project By Chaitanya Chobhe
</footer>

</body>
</html>
"""



def get_db():
    connection = sqlite3.connect(DATABASE)
    connection.execute("""
        CREATE TABLE IF NOT EXISTS students (
            roll INTEGER PRIMARY KEY,
            name TEXT NOT NULL,
            course TEXT NOT NULL,
            email TEXT
        )
    """)
    return connection


@app.route("/")
def home():
    query = request.args.get("q", "").strip()

    with get_db() as connection:
        if query:
            students = connection.execute("""
                SELECT roll, name, course, email
                FROM students
                WHERE CAST(roll AS TEXT) LIKE ?
                  OR name LIKE ?
```

```
            ORDER BY roll
        """, (f"%{query}%", f"%{query}%")).fetchall()
    else:
        students = connection.execute("""
            SELECT roll, name, course, email
            FROM students
            ORDER BY roll
        """).fetchall()

    return render_template_string(
        HTML, students=students, query=query
    )


@app.route("/add", methods=["POST"])
def add_student():
    roll = request.form["roll"]
    name = request.form["name"].strip()
    course = request.form["course"].strip()
    email = request.form.get("email", "").strip()

    if not name or not course:
        return "Name and course are required.", 400

    try:
        with get_db() as connection:
            connection.execute("""
                INSERT INTO students (roll, name, course, email)
                VALUES (?, ?, ?, ?)
            """, (roll, name, course, email))
    except sqlite3.IntegrityError:
        return "This roll number already exists. Go back and try again.", 400

    return redirect("/")


@app.route("/delete/<int:roll>", methods=["POST"])
def delete_student(roll):
    with get_db() as connection:
        connection.execute(
            "DELETE FROM students WHERE roll = ?", (roll,)
        )

    return redirect("/")


if __name__ == "__main__":
    port = int(os.environ.get("PORT", 5000))
    app.run(host="0.0.0.0", port=port, debug=False)
```